**Titanic Data Cleaning:**

First five records:

```
   survived  pclass     sex   age  sibsp  parch     fare  ...  class    who adult_male  deck
embark_town alive  alone
0         0       3    male  22.0      1      0   7.2500  ...  Third    man       True   NaN  Southampton
no  False
1         1       1  female  38.0      1      0  71.2833  ...  First  woman      False     C    Cherbourg
yes  False
2         1       3  female  26.0      0      0   7.9250  ...  Third  woman      False   NaN  Southampton
yes   True
3         1       1  female  35.0      1      0  53.1000  ...  First  woman      False     C  Southampton
yes  False
4         0       3    male  35.0      0      0   8.0500  ...  Third    man       True   NaN  Southampton
no   True

[5 rows x 15 columns]
```

Dataset shape: (891, 15)

Missing values:

```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

Shape after removing null rows: (182, 15)

Columns after removing null columns:

```
Index(['survived', 'pclass', 'sex', 'sibsp', 'parch', 'fare', 'class', 'who',
       'adult_male', 'alive', 'alone'],
      dtype='str')
```

Missing values after imputation:
survived 0
pclass 0
sex 0
age 0
sibsp 0
parch 0
fare 0
embarked 0
class 0
who 0
adult_male 0
deck 688
embark_town 0
alive 0
alone 0
dtype: int64

Duplicate rows: 110

Cleaned dataset:
survived Passenger_Class sex age sibsp parch ... deck embark_town alive alone survival_label fare_category
0 0 3 MALE 22.0 1 0 ... NaN Southampton no False Not Survived Low Fare
1 1 1 FEMALE 38.0 1 0 ... C Cherbourg yes False Survived High Fare
2 1 3 FEMALE 26.0 0 0 ... NaN Southampton yes True Survived Low Fare
3 1 1 FEMALE 35.0 1 0 ... C Southampton yes False Survived High Fare
4 0 3 MALE 35.0 0 0 ... NaN Southampton no True Not Survived Low Fare

[5 rows x 17 columns]

Final dataset shape: (781, 17)

Remaining missing values:
survived 0
Passenger_Class 0
sex 0
age 0

```
sibsp              0
parch              0
fare               0
embarked           0
class              0
who                0
adult_male         0
deck             579
embark_town        0
alive              0
alone              0
survival_label     0
fare_category      0
dtype: int64
```